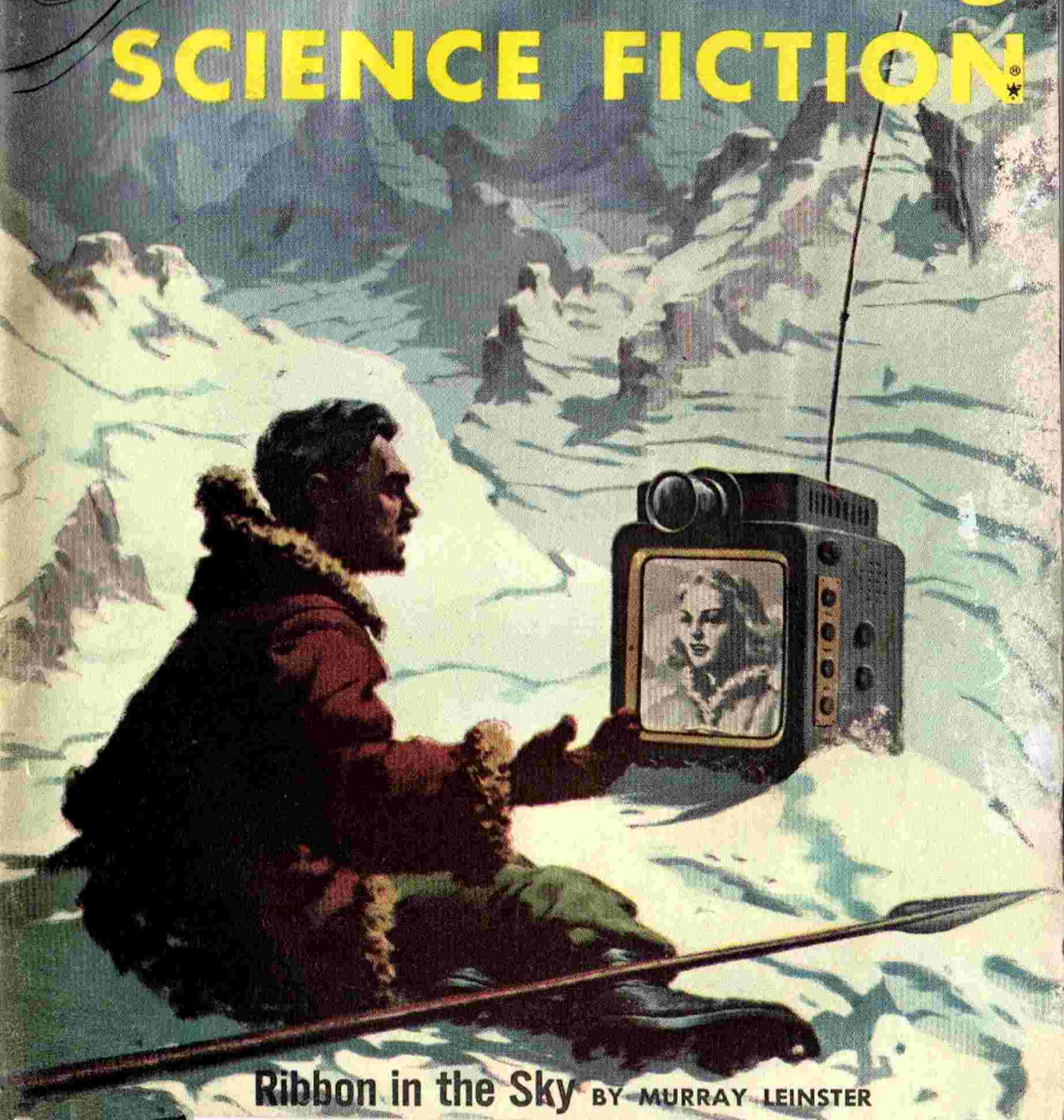


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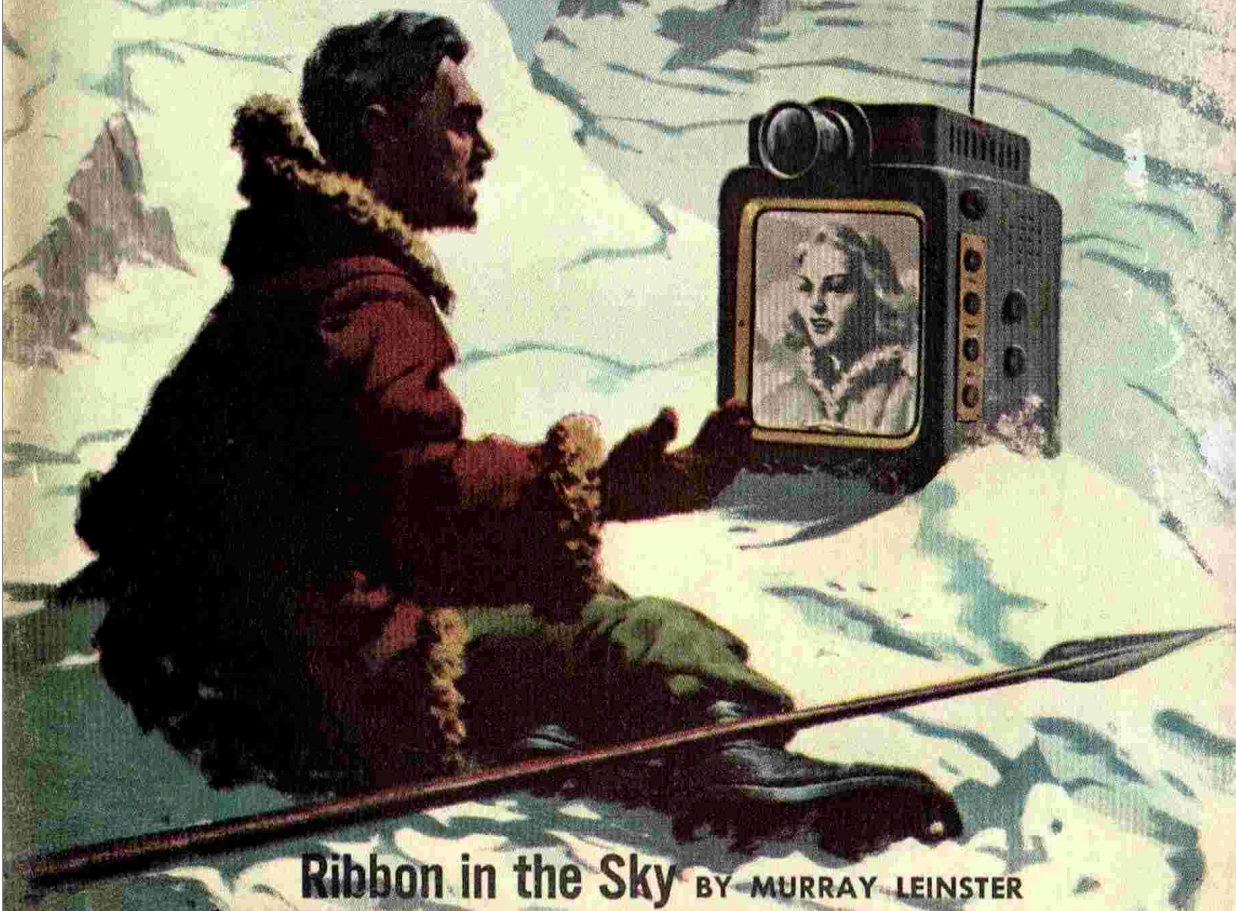
Astounding SCIENCE FICTION®



Ribbon in the Sky BY MURRAY LEINSTER

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*** START OF THE PROJECT GUTENBERG EBOOK NEEDLER ***

NEEDLER

BY RANDALL GARRETT

Illustrated by Ems

**"The principal difficulty in the case
... lay in the fact of there being too
much evidence. What was vital was
over-laid and hidden by what was
irrelevant."--Sherlock Holmes**

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They just didn't give a damn. The first load of survivors brought back after the Battle of Leymon's Star had been short-circuited somewhere, and they didn't give two hoots whether they lived or died.

The same thing happened to the crew of the GSS *Bedevin* after the skirmish in the Great Rift. The *Bedevin* was found drifting along, out of control, after having demolished an enemy vessel with a blast of the new *aJ* guns.

It was a case of "the operation was a success, but the doctor died." Or might as well have.

The crewmen of the fighting ships were in a state of semicatatonia.

The alien ships were burned and blasted out of space, with the exception of those which turned tail and ran. The survivors in the human ships were picked up and taken to Kandoris VI, the Galactic Main Base of the Interstellar Fleet.

Fleet Commander Allerdyce hospitalized the men and turned the problem over to the Civilian Research Corps. General Director Eckisster frowned over the whole mess, fired out assignments right and left, and dumped the bulk of the responsibility into the lap of Roysland Dwyn, chief of the Special Weapons Group.

Dwyn immediately asked for a specimen from the Fleet Hospital Psychiatric Ward.

Bilford, the chief psychometrist, brought one of the crew members from the *Bedevin* into the office of the head of Special Weapons four days after the survivors had been picked up.

Roysland Dwyn glanced up from the work at his desk when Bilford entered. Behind the huge plastic block of the desk, he looked no larger than the average man. It was only when he stood that it became apparent that Roysland Dwyn was two sizes larger than the average man, regardless of where you measured.



Bilford walked on into the office. "You wanted to see Captain Gisser, Roysland?"

Roysland nodded his massive head. "Bring him in; I want to get the whole picture on this business."

Bilford nodded and turned back toward the door. His eyes looked sad and pitying, and he ran a lean, nervous hand through his bushy gray hair as he called out: "All right, Captain Gisser—come in here."

As Captain Gisser strolled in from the outer office, Roysland watched him carefully.

Gisser was tall and graceful, in the near-perfect physical trim of a fighting man. He moved with military precision, but without the stiff rigidity of formal marching. He took one step through the door—and stopped.

Roysland narrowed his gray eyes and looked at the captain's face. The expression on it was definitely not the sleepy, glazed look of the hypnotic catatonic. After a moment, Roysland decided it could be described as a sort of apathetic introspection.

"How long will he stand like that?" he asked Bilford.

Bilford spread his hands. "Until someone tells him to move or he collapses from lack of food or sheer fatigue."

"Have him sit down over there." Roysland pointed. "No use making the poor guy stand up."

"Go over to that chair and sit down," Bilford told the captain. Gisser did as he was told.

Bilford pulled up another chair and sat down. "Why'd you want to see him?" he asked. "I mean, do you have anything in mind?"

Roysland shook his head. "Nothing specific; I'm just trying to see every angle of this. The Enlissa have a new weapon; we've got to do something to counteract it. So far, we don't know anything about it except that it bollixes up the brain—and that isn't very useful. It's like trying to deduce the existence of a pistol from the holes in the target."

"Worse," Bilford said gloomily; "we don't even have a hole to analyze."

"Yes, we do. A psychic hole." Roysland gestured toward the silent captain. "Are they all like that?"

"Essentially, yes," Bilford said.

"Can he hear what I'm saying? I mean, can he understand me?"

"That's a hard question to answer. I should say that the understanding was of a very low level. Here, I'll show you what I mean." He turned and looked directly at the seated spaceman.

"Captain Gisser, how old are you?" he asked in a firm, clear voice.

There was no answer.

"Gisser, when were you born?"

Still no answer.

"Gisser, *tell us* when you were born."

"Twelve, Eight, Seven sixty-four," Gisser said promptly.

Bilford looked back at Roysland. "He won't do anything on his own; there's absolutely no conscious volition. He has to be told what to do.

"Just asking him a question isn't enough; you have to insist on the answer. That's what I meant by saying that his understanding is on a very low level. He can't even deduce the presence of an unspoken command."

Roysland frowned and started to say something, but he was interrupted by a flicker of light on his desk panel.

He looked at Bilford. "The boss," he said dryly. Then he pressed a stud.

Light flickered in the air and coalesced into the seated figure of a portly, smiling, middle-aged man. The image wavered a little, then settled into an illusion of material solidity.

General Director Eckisster smiled and said: "Are we getting anywhere, gentlemen?"

"We're just getting started," Roysland said.

Eckisster nodded. "I see." His eyes lit on the captain, who was still sitting in the same position he had taken when he was ordered into the chair. "Is this one of the *Bedevin's* men?"

It was Bilford who answered. "Yes, sir. Captain Gisser, Prime Officer."

"And you haven't found out anything about him yet? Don't you know what's wrong with these men?" Eckisster's voice was bland on the surface, but there was a biting hardness underneath.

"We know what's wrong with them, sir," Bilford said stiffly; "we just don't know what caused it."

"According to the electroencephaloscope readings, the electrical activity of the prefrontal lobes is exhibiting a loop-feedback pattern. It's going around in circles without getting anywhere. As far as the nerve impulses are concerned, these men have been effectively lobotomized—almost completely so."

"I see." Eckisster looked at the captain again. "Captain, stand up." The captain stood. "Sit down." The officer sat. Eckisster rubbed a plump finger over his chin. "That's according to the report, at least. Would he kill himself if I asked him to?"

"Not if you *asked* him to," Bilford said coldly. "He might if you *told* him to. Do you want me to try it?"

"Don't be ridiculous!" the general director snapped. He looked at Roysland, who had been sitting quietly, waiting for Eckisster to finish. "Roysland, do you have any idea of the nature of this weapon?"

"None, sir," Roysland said quietly. "Neither I nor the psychologists have any idea what could do this to the human brain."

"Oh, no?" Eckisster's plump face smiled. "Haven't I heard something about microwaves at high intensity?"

Roysland nodded. "Sure. I know what you mean. But I was talking about doing it over a range of seven hundred million miles.

"We know that it can be done, but we don't know how the enemy did it. Look at it this way: If we'd found every one of these men with his skull bashed in, we could say that it had been done with a club. But that still wouldn't explain how it was done from better than a light-hour away."

"Besides," Bilford chipped in, "high intensity microwaves don't have that effect. They affect the brain, sure—but not that way."

Eckisster nodded and folded his hands placidly. "I understand. Well, gentlemen, I—" He stopped suddenly and looked to one side, out of the range of his pickup. A voice said: "This facsimile just came in on the ultrabeam, sir."



A hand materialized out of nowhere, holding a fac sheet; Eckisster took it, unfolded it, and read it. His eyes opened a trifle wider, and he looked up at Roysland.

"Roysland, they've used it again. The *Killiver* was picked up this side of the Noir Nebula, near Poulderr. They found her because of the automatic signals. Every man aboard was just like Captain Whatsisname, there. They're bringing the ship here, to Kandoris." He paused and looked at both men in turn. "If this keeps up," he said, "they'll have us whipped. It's your job to keep them from doing that. Now, you've got several trails to follow. Follow them, and get some answers; that's all."

His hand touched the arm rest of his chair, and abruptly the image dissolved into transparent air.

Bilford looked at Roysland. "I don't like the way he keeps needling people," he said. "It gets under my skin."

Roysland stood up. "He thinks that's the best way to get things done. Maybe it is; I really don't know. I do agree with him in one respect: we *have* to do something—what, I don't know, but something.

"We've been fighting the Enlissa for eighteen years. Up until last year, when we invented the *aJ* gun, there hadn't been an improvement on either side; they were winning because they had more ships.

"Then we get the *aJ* gun functioning, and use it against them; and when we do, it turns out that they have an even better weapon. I know what they mean when they say war is hell."

He stopped and looked at the captain. "Well, let's get on with it; I want to ask him a few questions."

Eighteen years of fighting hadn't seriously damaged either side, insofar as actual loss of life was concerned. Men in ships had been killed, of course, but no civilian had yet lost his life as a direct result of the Enlissa-Human war. The Enlissa hadn't gotten in close enough to occupied planets—yet.

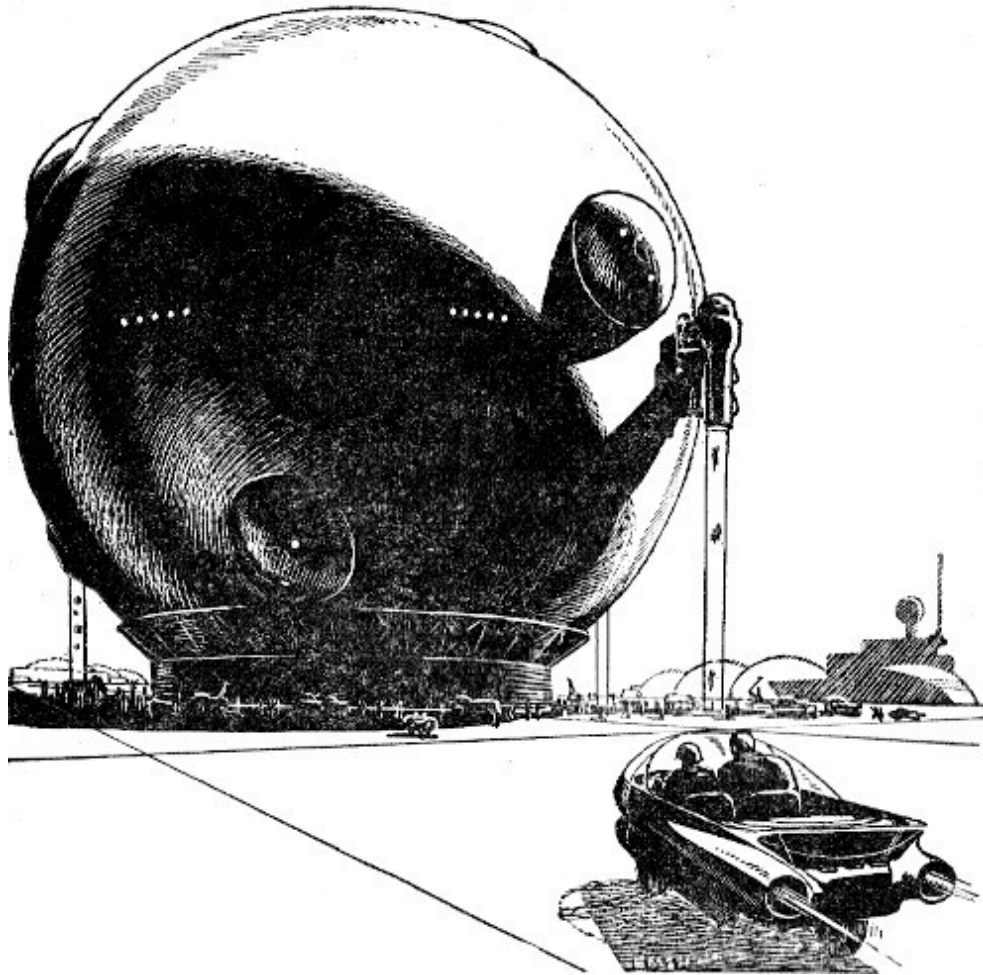
But, until a year ago, it had seemed inevitable that they would. The screen of ships that ranged around the periphery of the human-inhabited section of the galaxy was getting thinner all the time. The Enlissa had more ships, and, rather than make a direct attack, they seemed to prefer to punch at the screen, weakening it steadily.

But the Enlissa had underestimated human ingenuity. Both sides had been relying on the ultralight torpedoes to knock each other out of the sky, and humanity had realized that they had to have something better. So they had come up with the *aJ* projector. If matter can be projected through the no-space of ultralight velocities, why not energy?

The result was as devastating a heat beam as any dreamer could logically expect; all the energy of a nuclear reaction focused along a narrow locus of no-space toward the enemy ship. Even a shielded hull gives under bombardment like that.

It looked as though the war was won. That is, it did until ships came back with mindless crews.

The *Killiver* was sitting in its launching cradle at the far side of the ten-mile-square Grand Port of Kandoris. Roysland didn't bother to take the tubeway; he flashed his credentials and commandeered a surface jeep. Bilford had already taken charge of the crew, but Roysland wasn't worried about *them*; he wanted a look at the ship.



The *Killiver* was swarming with inspectors and special government investigators. Roysland jumped out of the jeep as it slowed near the giant sphere of the ship, and strode toward the ring of guards that surrounded the globe.

One of the guards looked up at Roysland's huge frame and said: "May I see your pass, sir?"

Roysland pulled out his pass and handed it to the guard.

The guard barely glanced at it; then he shook his head. "I'm sorry, sir; this is a general pass. You'll have to get one of the special passes for this ship. The Inspection Division has—"

"Where the devil do I get a pass?" Roysland snapped.

"You'll have to apply at Inspection," the guard said. "In person," he added.

Roysland shook his hand. "I'm not going twelve miles back to Administration. Who's in charge here?"

"Inspector Gowlan, sir."

"Call him; tell him Roysland Dwyn wants to see him."

The guard hesitated for a moment, then spoke softly into the communicator on his wrist. The speaker in his ear buzzed a reply. "He'll be right out," said the guard.

A moment later, a dark-haired, average-sized man in a chief inspector's uniform fell through the drop chute from the ship and crossed the open space toward Roysland. "Roysland Dwyn?" he said, holding out his hand. "You're Special Weapons, aren't you? I'm Gowlan."

Roysland nodded and gripped the proffered hand in his own great paw. "Glad to know you. I want to get on that ship."

The inspector shook his head. "'Fraid not ... not without a special pass. We've got to make damage estimates."

"That ship is equipped with *aJ* projectors," Roysland said. "My gang designed and built them from the ground up; I know more about them than you do. I want to see them—and the rest of the ship. I haven't got time to go gallivanting all over this base getting signatures on a blasted pass."

The inspector started to say something, but Roysland cut him off. "You can check with Eckisster, if you want; but hurry it up."

Gowlan looked up into Roysland's eyes, hesitated, then spoke into his wrist phone.

Less than two minutes later, Roysland was inside the ship.

The *Killiver* was in almost perfect shape. The *aJ* guns appeared to be in perfect operating condition, and the meters showed that three of them had tracked and fired at something that had passed the upper starboard quadrant of the vessel.

Roysland checked the recordings, then looked up at Gowlan, who had elected to follow him. "Any sign of the ship they were firing at?"

Gowlan shrugged. "The Space-fleet men didn't find anything. If the *Killiver* holed it, they would still probably be light-years away from where the ship was found."

"What made them skitter off like that?"

Gowlan looked at him. "I don't get it. What do you mean?"

Roysland waved his hand to indicate their surroundings. The corridors and rooms of the great ship were swarming with inspectors, who were photographing and checking every square centimeter of the ship.

"What happened? Why have we got this ship?" Roysland asked.

Gowlan thought for a moment, then nodded slowly. "I see what you're getting at. Let's see—

"The *Killiver* is cruising in ultradrive. They pick up a blip on the detector; it's an enemy ship. They're too far away to torpedo, but they're well within range of the *aJ* projectors. That gets us up to the moment of firing." He stopped and his frown deepened. "Wait a second; that doesn't make sense."

Roysland raised an eyebrow. "What doesn't?"

"Well, look here: The gunners would have had to be awake when the *aJ*'s were fired. All right; that means they tracked the Enlissa ship, then cut in the automatics to fire the *aJ*'s. They must have missed, because the Enlissa used the mindjammer *after* the *aJ*'s were fired.

"But if that's so, then why didn't the Enlissa ship capture the *Killiver*?"

It was a good point. Roysland frowned and turned the thing over in his mind. A spaceship is expensive—hellishly expensive; the cost of a fleet of seagoing battleships is nothing in comparison. So you don't waste ships, even the enemy's. The whole object of a space battle is to destroy the enemy

crew without destroying the ship. Even a badly-damaged interstellar vessel is worth saving.

The *Killiver* was in excellent condition. If the Enlissa ship were still in good shape after the battle, why hadn't they taken the *Killiver*?

"The only thing I can figure," Gowlan said, "is that the Enlissa ship fired their mindjammer just after the *aJ*'s were fired—almost at the same time, you might say." He grinned. "Sure. That's what must have happened."

Roysland nodded. "It looks like the only explanation," he agreed. "That is, except for one thing."

"What's that?" Gowlan wanted to know.

"Why has the same coincidence occurred in three different battles, in widely separated parts of the galaxy?"

Gowlan's face lost its self-satisfied look. "Yeah," he said softly. "Yeah. Why?"

"Kick that around a while," Roysland said, grinning. "If you come up with anything, let me know."

Roysland Dwyn spent the next two days sitting in his office with his feet on his desk, leaning back in a chair that creaked ominously with his weight. The only interruptions were for food and sleep—except when one of his staff called in with new data, which was rare.

He got one call from Milford. The microwave business that the general director suggested had shown some promise of snapping the stricken crews out of their apathy. Some of the men were improving rapidly, and others more slowly; but all of them were showing some positive response to the treatment.

On the afternoon of the second day, he got a call from Eckisster. The old man didn't look particularly jovial. His image solidified with a scowl on it. "What have you got on this microwave business?" he snapped.

Roysland lifted his big boots off the desk and leaned forward leisurely. "Nothing."

"You'd better get something fast," the general director said. "They're attacking shipping now, and they're well within the periphery."

Roysland jerked erect. "*What? What happened?*"

Eckisster's lower lip curled. "Don't use that tone of voice on me, Roysland. I don't like it. I want you to find out a few things. What's happening? Why do they attack this way and do nothing? What sort of gadget do they have? Is there any defense against it? Can we make it? Can we—"

His voice trailed off. Roysland had stood up and walked around his desk until he was less than a yard from the image of the general director. He knew full well that his own image in the director's office was doing the same thing. And in spite of the fact that Eckisster knew the image was harmless, Roysland's impressive mass quieted him.

When he spoke, Roysland's voice was low. "Now you listen to me, Eckisster. You want me to solve this problem. O.K. I want to figure it out as much as you do, but I can't do a thing without data. I have to know what has happened, and I have to know exactly how it happened. So don't come busting in on me with a lot of vague hints when I'm thinking. I don't have to put up with that sort of stuff; either give me the data on what happened, or go yak at someone else while I figure this out without any help from you."

Eckisster looked up at the bulk of bone and muscle that towered over him. "Don't get excited, Roysland," he said. "I'll forgive your impertinence: it's just that I'm so worried, myself."

"O.K. You're excused, too. Now, what's this about shipping being attacked?"

Eckisster glanced to one side and reached for something outside the pickup field. The end of his arm vanished and reappeared holding a sheaf of papers. "Of course, a copy of this will be sent to your office right away, but I can give you the essentials now.

"Two unarmed cargo vessels left Belixa III a week ago, bound for Niadel V. They were escorted by a light cruiser of the *Sidneg* class. They were picked up, off course, after they had passed the Niadel sun; nobody on board had even bothered to eat for four days.

"They probably wouldn't have been found at all if they'd been ordinary merchant vessels, but the local government on Niadel V was looking for them; there'd been an epidemic of some sort there, and these ships were on an emergency run with antibiotics of some kind."

Roysland stepped back and sat on the edge of his desk. "Got all three of them?"

"All three of them," said Eckisster emphatically. "Now, I'll send this report over to you immediately. We'll have to get some action. If the Enlissa can get in this close, they may decide to attack Kandoris itself! Your job is Special Weapons. Find a screen of some sort that will protect us from this—whatever it is."

"Call it a mindjammer," Roysland said. "One of the inspectors used that word, and I kind of like it."

"You like it." Eckisster's voice was cutting. "I don't like anything that does *that* to a human brain. Get busy and find some way to beat it."

Roysland started to explain that he liked the word—not the object—but the general director's image was already dissolving. Roysland stepped back behind his desk and dialed a number. A few seconds later, Bilford's image materialized. The nervous little man looked more nervous than ever.

"What is it, Roysland? More trouble? I hope not. I've had Eckisster on my neck all morning."

"I know; I just got him off mine. But I wanted to ask you something. Is there any correlation between the frequencies that help those men and the frequency of the feedback circuit in their prefrontal lobes?"

Bilford frowned in thought. "I don't know; I'd never thought of it from that angle. They don't have any obvious correlation, I can tell you that. I'll check on it, though. I'll run it through the differential analyzer."

Roysland nodded. "Try that. Let me know if you get anything."

He cut Bilford off and dialed another number. The image that appeared this time was wearing the uniform of a fleet commander.

"Commander Allergyce, do you mind if I ask you a couple of questions?"

"Go ahead, Roysland. What is it? I hope you're not going to needle me the way your boss does. I'd have tossed him out of my office, except that you can't grab a solidiphone image. Best I could do was shut him off, which was very unsatisfactory." The commander grinned wryly at the thought.

A big grin spread itself across Roysland's blocky face. "I know how you feel. No, commander, I just wanted to ask a couple of questions, as I said.

"You're familiar with the details of the Enlissa attack on that medical supply convoy?"

The fleet commander nodded.

"Well," Roysland continued, "what would happen if you were in command of the cruiser and you found a trace on the scope that indicated an Enlissa ship?"

"The orders cover that," said the commander. "The cruiser cuts in with the *aJ* guns before the Enlissa ship gets within torpedo range."

"And this always works?"

The commander shrugged. "It always has so far, the *aJ*'s knock them out of space before they can get close enough to launch screenbuster torpedoes accurately. But this new gadget they've got evidently has as great a range as the *aJ* projectors."

"Or greater," Roysland added.

"Yeah," said Allergyce softly, "or greater."

"Is there any other possibility?" Roysland wanted to know.

The commander nodded. "One—if the Enlissa were lucky, that is. If the enemy ship could have approached the convoy by coming in directly from a star, the subetheric radiation from the sun behind them would blank out their own radiation, and they could get in pretty close before they registered on a screen.

"But in order to do that, they'd have to know the convoy's course and lie in wait for it. If they actually did use a star to hide themselves, it was probably pure luck on their part that they happened to be in the right place at the right time."

Roysland nodded slowly, his eyes narrowed in thought. "I wonder—" he said finally. "Would you do me a favor, commander? Would you check and see if that cruiser actually fired towards a sun? That might give us some information."

"I'll check," Allerdyce said. "It'll be on the recorders. I'll let you know what I come up with."

"Fine," said Roysland. "I'll see you later." He cut off, and his image disintegrated.

Roysland looked at the dark, blurred reflection of his face in the black plastic of his desk for a moment, then grinned. "All right, buster," he said to the face in the desk, "you're stuck for a while, anyway. Time to call in some help."

He touched a switch plate on his desk panel and said: "Call a meeting of the Special Weapons Staff at my home at twenty-nine hundred hours."

He touched another plate and said: "As soon as the report from the general director comes in, have it transferred to my home."

And another: "Send all data to date on the enemy's latest weapon to my home. Code it *Mindjammer*."

Then he got up, shut off his desk, and went out. An early meal was on the agenda, it seemed.

Blackpool's Restaurant was, as usual, well populated, but not over-crowded. Roysland managed to find a table in the rear, where he sat down and ordered a tall glass of fruit juice. He liked Blackpool's; its old-fashioned, almost primitive atmosphere was impressive without being phony. The waiters—remote-control humanoids guided by the vast robot brain in the basement—were dressed in the fluffy, bright, fluorescent clothing of a style that had been worn two centuries before, when Blackpool's had been built. The uniforms had never changed.

Roysland consulted the menu, told the waiter what he wanted, and went back to his fruit juice.

"Roysland? Mind if I pull in?"

Roysland looked up at the short, round-faced, smiling man standing by the table. "Not at all Osteban; sit down." Roysland didn't particularly want to

talk to him then, but it wouldn't do to offend the Galactic News Service. Roysland waved the man to a seat and asked him if he wanted a drink.

Osteban eyed his host's drink. "What are you drinking? Want to let me taste it?" He took the glass, sipped at it, and made a wry face. "F'revvinsake! Mind if I have something with life in it?"

Roysland said he didn't, and Osteban ordered something more potent. When the waiter brought it, he took a healthy swallow and then said: "Mind if I ask a question?"

"Ask to your heart's content," Roysland said. "You will, anyway. But I don't guarantee any answers."

"Did you ever?" He took another swallow of liquid. "What's in this rumor that the Enlissa have invented a gadget that drives people crazy?"

"I haven't heard any such rumor," Roysland said. It was a perfectly true statement, if a trifle incomplete.

"Did I ask if you'd heard it?" Osteban countered.

"Tell me something, Osteban," Roysland said seriously. "Did you ever use a declarative sentence in your life?"

"What do you mean? Let's quit the kidding, shall we? Didn't you understand my question—or are you playing dumb?" Osteban grinned as he said it, making it totally inoffensive.

Roysland flipped a coin, mentally. It came down tails, and Osteban lost. "I can't speak officially, of course," Roysland said, "I'll just have to be a 'reliable anonymous source.' But I can tell you this: We don't know what the Enlissa may or may not have; but we haven't lost any ships because of any insanity rays, or what have you."

"Is that a fact?" Osteban thought for a moment. "I guess it is or you wouldn't say it, would you?"

They drank in silence for a few moments, then Osteban said: "All right, tell me something else, will you? These new *aJ* projectors have been on active duty for half a year or so, haven't they? They're supposed to be hot stuff, right? Then why is it that they haven't destroyed any enemy ships? Why is it that all the communiques always say: 'The Enlissa ship was finally destroyed by ultralight torpedoes.'"

Roysland frowned. "I didn't know that was the case. However, I think I can hazard a guess. An *aJ* projector requires the installation of a big no-space generator, similar to the one that drives the ship. They're expensive when they get that big, and only a few of the larger battleships have been equipped with them.

"Now, actually, what are the odds that any particular ship will make shooting contact with the enemy? Very small. The probable reason that no enemy ships have been destroyed by *aJ* projectors is that no *aJ* ships have come in contact with the enemy."

"Do you think that's it?" The reporter grinned and took a final sip from his glass, draining it. "Well, I guess I can't get a story out of you, can I? O.K., then; will I see you around?"

"Sure," said Roysland. "Take it easy."

But the reporter had ruined his dinner. What was there about the casualty statistics that was unusual? Was there any more information in that area? He'd have to check and see.

The executive staff of Special Weapons assembled in Roysland Dwyn's study via solidiphone at 2900 that evening. There were five of them at the table. Kiffer, Mardis, Taddibol, and Vanisson were actually thousands of parsecs away, on four widely scattered bases of the fleet.

Roysland Dwyn, himself, was the fifth man.

"I'm going to make this short and sweet," Roysland said. "I don't want much discussion until you've all had a chance to mull over the data in your minds for a while."

He spent fifteen minutes telling them what he'd picked up so far. When he was finished, Vanisson asked: "Have you tried running this through a computer?"

Roysland shook his head. "It can't be done. We don't have enough symbolizable data. Only the human mind can take incomplete data and come

up with the right answer; we're going to have to do this ourselves. We'll have to probe into what we have and see if we come up with anything."

"I've got a question," Mardis said. "Why does the enemy only pick on *aJ* ships?"

Roysland nodded. "And why do they invariably fire *immediately* after the *aJ* projectors fire?"

Kiffer said: "Could it be some kind of subetheric vibration that does the trick?"

"You're the subelectronics man," Roysland said. "What do you think?"

Kiffer shrugged. "Subetherics are dangerous; near a projector, they can foul up electrical currents, provided the currents aren't too strong. They can knock a man out, or even kill him; but I never heard of any effect like this."

"What would it take to get an effect like this?" Roysland asked. "Figure it from that angle."

Taddibol looked excited. "Could it be that the enemy doesn't even have such a weapon?"

They all looked at him. Roysland was grinning. "Maybe you've got the same hunch I have," Roysland said. "Let's hear it."

"We know: one, it only happens on *aJ* ships; two, it happens at the instant of firing. Could it be some sort of backlash from the projectors that's doing it?"

Roysland, still grinning, looked at the subelectronics man. "How about it, Kiffer?"

Kiffer shook his head. "I doubt it. There's a backwash, of course, as there is to any kind of no-space generator. But it's almost undetectable, even with subelectronic instruments. There's certainly not enough to hurt anyone. Besides, the emission would be from the exciter in the gun, and it would hit the men in one direction; that might slow their neural currents up a little for a fraction of a second, but it wouldn't do anything like what we have here, even if it were strong enough."

All the time he had been talking, Mardis had been nodding his head in agreement. When Kiffer finished, Mardis said: "And besides that, we've

tested the things, remember? We fired those projectors under every condition we could think of, and we didn't get any feedback lobotomies."

Taddibol nodded. "That's right. We mounted four projectors on the X-69, and melted asteroids for six months before we released the weapon to the fleet."

"Anybody got any more questions?" Roysland asked.

There were none.

"All right, I have some I want you to think over. First: Is this really an enemy weapon? Second: If so, how is it generated and projected at *aJ* ships? Third: If it isn't an enemy weapon, what is it? Fourth: Regardless of what it is, *where* is it generated? Fifth: If we—"

He didn't finish. The solidiphone signal was blinking. He activated the instrument, and Eckisster coalesced into the room, his chubby face dewy with perspiration.

"Ah!" he said. "I'm glad to find you at home. I'm glad to see you're working on this thing at last. Why didn't you call in your staff two days ago? Maybe they can figure something out, even if you can't; this thing has suddenly become dangerous."

Roysland looked dangerous, so the general director patted the air with a hand. "I've got the stuff for you right here, Roysland, so don't give me any of your lip. In the first place, there was a convoy attack yesterday out near the periphery. It turned out to be one of the biggest battles of the war so far. The enemy lost five ships to fire from *aJ* projectors, and four to torpedoes. We lost two ships to torpedo fire and six ships to the ... what did you call it? ... mindjammer.

"Fortunately, we had them out-numbered and were able to recover the crews and ships we'd lost to the mindjammer.

"But it doesn't look good. If they start using that weapon on a big scale, we'll be sunk. If they ever hit a planet with it—Well, you can imagine what it would be like to take care of a city full of morons."

Eckisster paused, squinted his eyes at Roysland, and jabbed at him with a finger. "Now, I've got an idea," he said. "We've got to develop some sort of screen that will take care of the mindjamming effect. You ought to be pretty good at defensive screening by now; until you worked out the *aJ* projector, Special Weapons has been strictly on the defensive side."

Vanisson said: "Naturally, sir. It's easier to prevent something from getting to you than to figure out a way of getting to the other guy. Arms theory shows —"

Eckisster glowered at the man. "Theory, hogwash! I want a defense against the mindjammer, and I want it yesterday! Get busy!"

Roysland was leaning back in his chair with his arms folded over his chest. When Eckisster had completed his outburst, Roysland said, calmly: "Are you quite through, sir?"

"I am," said the general director. "I doubt if you mudheads can come up with anything before we are all reduced to gibbering idiots, but God knows I've done my best."

"You *are* finished then?" Roysland's voice was still calm. Then, quite suddenly, it became savage. "Then leave us alone, so we can think! Good-by!" He snapped off his receiver switch, and Eckisster's image vanished before the director had a chance to say anything.

Roysland smiled gently. "And now, gentlemen, let's get down to work."

Two days later, the X-69—the fast, experimental ship of Special Weapons—dropped down to the Grand Port of Kandoris. A score of heavy trucks, loaded with equipment, waited for the cargo ports to open; and big, lumbering sections of construction framework were being moved in toward it.

The man who floated down the drop chute from the equatorial air lock was Kiffer Samm. A ground taxi was waiting for him, and it started to move even before Kiffer closed the door.



Within minutes, he was in Roysland Dwyn's office. He pulled up a chair, sat down, and said: "Well, I'm here."

"An astute observation," said Roysland. "Who knows to what depths of scientific thought you may reach with such cosmos-shaking revelations as that?"

"A mere nothing," said Kiffer; "I might add that the X-69 is here, too. How long will it take to get the stuff mounted on her?"

"A couple of hours. I made sure that Allerdyce would have the necessary equipment ready when you landed. We'll take off as soon as she's loaded."

Kiffer frowned at Roysland, then looked down at his fingernails. "You don't need to go along."

"Why not?"

Kiffer kept looking at his nails for a full five seconds. Then he looked up and said: "Look, Roysland, suppose what you suspect *is* true. Suppose that it isn't an enemy weapon, but a backfire from the *aJ* guns. If so, then we'll be mindjammed when we test out the fleet's weapons. And we can't afford to have you in that condition."

"I know it," said Roysland, "but there's no other way I can get the data. Besides, Bilford is having some success with using microwaves on the patients; there's reason to believe that the condition is temporary."

Kiffer shrugged and spread his hands. "O.K.; if that's your orders—" He let his voice trail off. Then: "But I still don't like it. Look at it from my viewpoint; if I'm knocked out, I can depend on you to figure out a way to bring me out of it. But if you're out, too, what's to become of me?"

Roysland laughed. "That's the best reason you could have given. Thanks. But I'm still going."

It took just a little more than two hours for the Space-fleet ordnance crews to replace the *aJ* projectors on the X-69. Roysland's theory was simple. Although the *aJ* guns might be responsible for the mindjamming effect, it was obvious that they didn't cause it every time. It was possible that there were slight differences in the backwash of radiation—slight differences caused by variations in the projectors themselves. The weapons of the *Bedevin* and the *Killiver* went into the turrets of the X-69; if there were any basis for the theory, at least two of those guns would be responsible for the mindjamming effect.

The X-69 left Kandoris VI at 0500 hours, aimed herself for the vast void of the Lesser Rift, and cut in her no-space generators. The drive slammed her abruptly up past the velocity of light and into multiples thereof.

Roysland had a cabin to himself near the upper deck at the nose of the ship, just beneath the control bridge. With Kiffer's aid, he set up recording instruments at various points throughout the ship, started them, and promptly forgot them. He was aboard as a human observer; the instruments had their own job to do.

Roysland pushed his muscular bulk up the stair to the control bridge. Above him rose the hard, transparent dome of the ship's nose. He stood for a moment, watching the stars move slowly by. Then he walked over to where Kiffer and the ship's officers were standing, near the main control area.

"Captain Dobrin," he said, "we've got our instruments set up; we'd like to find some targets to test-fire at." He paused for a moment and looked at the officer. "You know what we're up against, don't you?"

Captain Dobrin was a lean, graying, grim-faced man who looked as though the last time he had smiled was in his mother's arms. "I know what our chances are; slightly worse than those of a fighting ship engaging the enemy, as I figure it. Besides, I figure that if you're willing to risk your neck—or your mind—I'll take the same chances with the ship." He stopped and looked at the screen, then looked straight up, pointing his finger through the transparent dome of the nose. "We'll head toward that star, there; it's a triple sun, and there's usually plenty of debris floating around in the vicinity of a system like that."

Roysland watched as the ship approached the triple star system. At first it was only a bright point of light. Then, gradually, it separated into two lights, one several times as bright as the other. Finally, the brighter of the two separated into two parts. The three suns stood at the points of an elongated isosceles triangle.

As they neared the trio, the captain ordered the no-space generator cut, and the ship dropped out of drive. Instead of having a velocity measured in light-hours per second, the ship dropped suddenly to miles per second.

"Electromagnetic detectors on," said the Fire Control Officer.

A ship traveling above the velocity of light cannot detect a material body unless there is subetheric radiation coming from the detected body. A star, naturally, can be detected. At those velocities, a star's subetheric radiation can be seen as ordinary light. But there is no way to detect a nonradiating body; in order to fire at a target, it's necessary to cut out the drive and use ordinary detectors to find a nonradiating body such as a meteorite.

"Target at forty million miles," said an observer.

"Track and fire," said the fire control officer.

The robot-controlled *aJ* projectors swiveled in their mounts, found the mass of nickel-iron that was their target, and hummed softly. Then they clicked.

That was all. Roysland neither saw nor felt anything unusual.

Three and a half minutes later, tardy light brought the news that the meteorite had flared in an actinic blaze of incandescent gas.

"Dead hit," said the observer.

Captain Dobrin looked at Roysland with a silent question.

Roysland nodded. "Go ahead. Let's pick out a few more; let's burn asteroids for a while."

They blasted eighteen planetoids into flaming gas in the next three hours. Roysland Dwyn and Kiffer Samm checked their instrument recordings and ran them through the differential analyzer after each firing.

"There's backwash, of course," said Kiffer. He pointed at a line that wavered up and down near the bottom of the graph. "That's the background—stellar noise from the subelectronic radiation of the nearby stars. Now"—he moved his finger along the graph—"this is the harmonic set up by the backwash at the instant of firing of the *aJ* projectors.

"It looks pretty high on the graph, but that's because the subnuclear reactions inside a star are so slight that they don't generate much background noise. Actually, the backwash from the *aJ*'s couldn't possibly be called dangerous."

Roysland frowned; his heavy, dark brows pulled down, wrinkling his massive forehead. "Well, they obviously didn't do anything to us. At least, if they did, I haven't noticed it."

Kiffer shrugged. "Nothing harmful, anyway. Now, here's some comparison charts I have; the test runs on *aJ* guns that have been installed in other ships. The wave form is identical; these guns don't react any differently than any other. As far as I can see, there's no reason for these guns to have knocked out the crews of those ships."

Roysland rubbed a finger across his chin and stared at the ceiling. That chin-rubbing gesture was significant to Kiffer; he knew Roysland well enough to know that the big man was thinking. Kiffer kept his mouth shut and waited.

Finally, Roysland snapped his fingers. "Look," he said sharply, "why aren't these things tested the way they're used?"

Kiffer looked puzzled. "The way they're used?" He paused a moment. "Oh, I see what you mean. Why aren't they test-fired while the ship is in no-space drive? That's easy. They have to be connected up to the trackers, and the trackers can't fire at undetectable objects. And you can't detect a meteorite in no-space drive.

"Of course, I suppose we could send out some torpedoes and try to hit them, but that would be sort of wasteful."

"Then the guns aren't tested in no-space, huh?" Roysland said grinning. "Then somebody's been falsifying reports to my office."

Kiffer grinned back. "Sure," he said, "they're tested, but without the robot trackers; I don't see what difference that would make, though."

"Let's not jump to any conclusions. Those things fire in sequence when they're tracking—one right after another, in battery. And they're timed so close together that they might as well be going off all at once. Or, the time lag may have something to do with it, short as it is. Suppose we fire them in no-space drive, just as if it were battle conditions."

"At what?" Kiffer wanted to know. "The robot can't track unless it has a target."

"We've got targets," Roysland said quietly. "Millions of 'em."

"The Torpedoes? But—wait a minute! Millions?" Kiffer slapped his palm against his forehead. "Why didn't I think of it before? The stars, of course!"

"Right," said Roysland. "They radiate in the subetherics. But no one ever thought of firing at them before, because there's no way of telling whether you hit it or not; a star could soak up all the energy of the whole Galactic Fleet without noticing it. But we don't care whether we *hit* the target or not; all we want is a target to *fire* at."

"I'll reset the recorders," Kiffer said. "Let's see what happens."

"I'm going up to the bridge," Roysland said. "Set those gimmicks going; we want a record, even if this knocks us silly."

Up on the bridge, Roysland explained what he wanted done to Captain Dobrin.

"It can't hurt anything," Dobrin said. "We'll take a pot shot at the dwarf out there. They give a fairly small, bright target."

The ship plunged into the no-space of ultradrive as the generators were cut in, and she began to move toward a point just to one side of the dimmest of the three stars.

"Target at three fifty-two million," said the observer.

"Track and fire," said the FCO.

Roysland held his breath as the projectors moved, hummed, and clicked again. And nothing happened.

Roysland let his breath out slowly.

"Was that O.K.?" the fire control officer asked. "We can't tell whether we hit or not."

"I doubt if you could miss even a white dwarf star at this range," Roysland said. "But you're right, of course; there's no way of being *absolutely* positive." He turned back to the captain. "Let's play around with this for a while. Make a few passes, back and forth at that star and let's see what we get on the recorders."

What they got didn't look like much.

"Here's the background noise," Kiffer said, pointing at the graph. "This time, it's almost a perfect sine wave; it's the backwash from the drive generators. Here's the harmonic generated when the *aJ*'s go off. And here"—he pulled a strip from the differential analyzer—"are the components. This one is the container phase for the energy envelope that holds the raw violence of the beam itself. And this is the carrier wave phase."

Roysland looked at the graphs and shook his head slowly. "And it all looks perfectly harmless."

"Looks, hell!" said Kiffer. "It *is* harmless. Believe me, Roysland, it is definitely not the backwash from the *aJ* guns that's causing the mindjammer effect. We'll have to look somewhere else."

"I guess you're right," Roysland agreed reluctantly. "If it isn't here—" His voice trailed off. He was right back where he started, and he didn't have anything to go on. Finally, he reached over to the intercom and punched for the bridge. "O.K., captain," he said, "let's turn the thing around and go home!"

Two weeks after the X-69 landed at Grand Base, Roysland still was stewing around, trying to make sense out of all the data he had.

Report from Bilford Vell, Chief Psychometrist: "The patients seem to be responding fairly well under the microwave treatment. It seems to act very similarly to the electro-shock treatments reputedly used centuries ago for certain types of insanity, although without the deleterious effects. The feedback loop in the prefrontal lobes is partially canceled out when the frequencies of the cerebral activities are the same as, and ninety degrees out of phase with, the microwaves beamed at the head.

"Naturally, this means that a series of treatments is necessary, since the cerebral frequencies are unpredictable and variable, and since the currents in the feedback loop are composed of a number of different frequencies."

Fine, thought Roysland. *There's some hope, at least. We know what can cure it, but what can cause it?*

Report from Allerdyce Blyt, CinC, Galactic Fleet: "I don't know what you can make out of this, but maybe you can get together with Bilford and figure out what it means. If you ask me, I think the Enlissa have gone nuts. Is it possible there's a backwash from their mindjammer?"

"Anyway, here's what's happened.

"During a minor skirmish near the Alavard Cluster, two Enlissa ships came in on attack geodesics toward the GSS *Viwil*. The *Viwil* is not equipped with *aJ* projectors, so they had to rely on conventional torpedoes. Since the odds were two to one, they had little hope of surviving, but they had hopes of

inflicting some damage on the enemy. So they waited until the Enlissa ships were well within range, and fired.

"The Enlissa ships took no evasive action, and the torpedoes destroyed both ships. There was no need for the *Viwil* to use evasive action, since the enemy ships *did not fire a single torpedo!*

"There have been other instances of similar action.

"In other small skirmishes, the *aJ* guns have proven their effectiveness; they've shot up Enlissa ships before they were in torpedo range. Oddly enough, no human ship equipped with *aJ*'s has ever been hit by a torpedo."

Roysland went back and reread one of the sentences. "Is it possible there's a backwash from their mindjammer?"

It's possible, sure. Until we know what the mindjammer is, we'll have to admit that anything's possible.

Report from Kiffer Samm: "I've done the checking you suggested. There is a definite effect on the brain, but it isn't permanent, nor noticeable. The backwash of the *aJ* guns causes a slight retardation of nerve impulses. But it isn't enough to cause any reaction—either mental or physiological. It doesn't last enough, in the first place; and it isn't powerful enough, in the second. I don't know what would happen if a person were subjected to such a field over a long period of time, but the situation corrects itself so rapidly that there is no danger of cumulative effects.

"Besides, some of the men affected have never been exposed to the backwash from *aJ* fire before, while others have been exposed a good many times. If the thing were cumulative, we would have men being knocked out here and there, at random, as the accumulation built up—and it just ain't so.

"The only parallel I can make—as far as long-range effects are concerned—are the effects of the backwash from the drive itself. And that isn't bad at all. Statistically speaking, the crews of spaceships are more alert, and have more interest in their surroundings, *after* long periods of service than they have before exposure. Even so, that is probably due to military work and periodic psychological checkups, rather than to any effect of the field.

"Do you have any other ideas?"

Roysland looked sourly at the report. *Ideas? Sure; I've got all kinds of ideas. I wish I had an answer.*

Report from General Director Eckisster—delivered via solidiphone:

"Roysland, you're going to have to start moving, here!" The director shifted his heavy bulk in his chair and glowered at Roysland Dwyn. "As far as I can tell, you haven't done a blasted thing! Of all the meaningless reports I ever read, these are the epitome of nonsense." He waved a chubby hand at a pile of papers in his lap. "As I understand it, you've been looking for some sort of effect emanating from our own weapons instead of from the enemy's."

"Now, to me, that's as silly as a man with a sword trying to explain away the stab wound in his belly by claiming that something happened during the fight and the hilt stabbed him. Or a man with a bullet wound trying to claim it was caused by the recoil of his blunderbuss!"

Roysland tapped his fingers softly on the top of his huge black desk until Eckisster was through, then he said: "It's the only hypothesis that fits the facts. I'll admit that we haven't been able to prove anything yet, but I'm convinced that—"

He was interrupted by the chiming of the solidiphone. He cut in a second circuit, and Fleet Commander Allerdyce coalesced in the air next to Eckisster. He glanced at the general director.

"Good afternoon, Eckisster." Then he looked back at Roysland. "I've got your weapon for you. Forty hours ago, Squadron 8477 met the enemy near St. Jairus' Cluster. We won the battle by a small margin, but that's neither here nor there. The important thing right now is what the hospital and salvage ships found when they came in after the battle. All the data isn't in yet, but as near as we can tell so far, a freak accident occurred.

"One of our ships was surprised by an Enlissa ship that came in out of a nearby sun; the enemy ship actually snapped by at less than a hundred miles. A lucky shot hit the drive generators of the enemy ship, and it stopped almost dead in space.

"They managed to get the crew of our own ship with their mindjammer, but something happened aboard the enemy ship, too. Evidently the weapon does have a backwash; the enemy crew was mindjammed, too!"

Roysland and Eckisster both started to say something, but the commander raised his hand. "Wait a second! The point I'm getting at is this: The Enlissa ship was recovered intact; the mindjammer projectors are aboard! I've sent an emergency order to the squadron commander in that sector; the Enlissa ship will be here tomorrow morning. We'll hold it sealed until you and your crew can investigate. The inspectors will have to go in with you, of course, but you'll be in charge of the weapons themselves."

He stopped and speared Eckisster with a frosty look. "I trust that meets with your approval, Eckisster?"

The general director was beaming seraphically. "It does, commander; indeed it does. Thank you. Thank you, so much."

Allerdyce glowered. "I'll be available in a couple of hours. Right now, I've got to get some work done." He cut the circuit.

Eckisster turned his beaming visage from the dissolving image of Allerdyce to the blocky figure of Roysland.

"May I suggest that you try investigating what few facts the fleet may have turned up? Who knows—you may find them profitable, eh? Or perhaps you're too busy trying to figure out how the *aJ* guns work to have any time for the enemy's mindjammer?"

"However that may be, I'll leave you to your work, bumblehead."

Roysland shot to his feet. "Good! Maybe I could get some work done, myself, if you weren't around needling me!" He reached out to snap off the solidiphone switch, but Eckisster, still smiling benevolently, was already fading. Roysland got the impression that his smile, Cheshirelike, still lingered after he had gone.

The crew of the Enlissa ship were the first live aliens ever seen by human beings. Their corpses had been dissected by the thousands, but the living

organism had never been investigated before.

"This gives us a jump on them," one of the biologists said. "As far as we know, no living human has ever been caught by the Enlissa."

Roysland, who was watching the aliens being herded out of the captured ship, turned his head to look at the biologist. "They don't know we've got this ship, either," he said.

The biologist blinked, then nodded. "Yeah. I see what you mean."

They were standing on the broad spread of plastalloy that covered the great landing field of Grand Base, standing in the shadow of the huge alien ship. The Psych men were pushing the Enlissa out of the ship, through the path formed by the Inspection Corps men and Roysland's own Special Weapons Group of the Research Division. The Psych men simply pushed them into the drop chutes from the ship. Other Psych men kept them moving toward the trucks that were taking them away.

The Enlissa weren't quite as tall, on the average, as a human being. The skeletal structure was a little heavier, and the section corresponding to the human rib cage was a series of armor plates that completely enclosed the viscera. The pale blue-violet of their skins came from the cobalt-protein complex that carried the oxygen through their blood, performing the same function that hemoglobin does in the human animal.



They were noseless; breathing was done through the mouth. The teeth were widely spaced, and the lips could not close over them, thus allowing the Enlissa to breathe, even when unconscious. The eyes were a solid black. It was impossible to tell, from a superficial inspection, where the deeply-pigmented surface of the eyeball ended and the dead black of the lens opening began. They were somewhat larger than human eyes, but they were set in front of the skull, allowing stereoscopic vision.

Their protective covering might have been called hair, by stretching the definition somewhat. By an equal amount of stretching, it could have been called fingernails or scales. It would have taken an awful lot of stretching to call it feathers.

The "hair" consisted of ribbons of thin chitinlike material. The ribbons weren't much thicker than human hair, but they were nearly a sixteenth of an

inch in width, and ranged in color from a glossy black to a royal blue, depending on the individual.

The feet were splayed, almost radial; the hands were four-digited—double thumbed and double fingered.

The clothing they wore, though radically cut, was analogous to the styles worn by human beings.

Roysland waited until the aliens were herded out of the ship. They had to be prodded like beasts, since there was no way to talk to them. No exchange of language had ever been achieved; but, like their human counterparts, the mindjammed Enlissa seemed to be perfectly willing to obey any exterior commands.

"What?" said Roysland. He had been so engrossed in his own thoughts that he had only dimly realized that Kiffer Samm was talking to him.

"I said that we'll have to check on them, too, after we see what this weapon is all about."

Roysland folded his hands and rubbed his thumbs together. "Maybe before."

"Huh?"

"Never mind," Roysland said. "Here come the last of them. We want to get all the samples out of their supplies that we can, and we've already been promised first look at those projectors the Enlissa have on board the ship. Come on; let's take a look."

The Enlissa ship wasn't organized too differently from the human version. On the surface, things looked odd; but the laws of the universe function the same way in all places, so the internal workings of the ship were essentially similar.

The Special Weapons men went through the ship with the men of the Inspection Division, photographing, tracing circuits, analyzing, checking differences, and organizing similarities.

Roysland and Kiffer spent most of their time with the big, complex projectors that were cradled in the hull blisters.

When Kiffer first saw them, he turned to Roysland and tried to keep from looking bewildered. "They're subelectronic projectors of some kind. But

what kind?"

"That's what we've got to find out," Roysland told him. "We'll have to find out what they do on a physical level first. From there, we'll go on to the physiological level; then we may—just *may*—be able to go on to the psychological effects."

Kiffer Samm looked up at the great frame of his superior, and grinned sardonically. "O.K. Now we've got the effect and the weapon that causes it. Can we correlate the two?"

Roysland shrugged his broad shoulders. "Sure we can. But how long will it take us?"

The laws of the universe may not differ from place to place, but the methods of using them do; and the particular laws that may be discovered in one place aren't necessarily the same ones that are discovered in another. No two human beings think alike; two different evolutionary branches of intelligence, stemming from totally different beginnings, certainly can't be expected to reason similarly. The amazing thing about the Enlissa was not the ways in which they differed from humanity, but the ways in which they were similar.

So it wasn't to be wondered at that the Special Weapons technicians couldn't figure out for the life of them what the projectors from the Enlissa ship did, or why they worked. If they had been the type of men to be stymied by seemingly-unbreakable barriers, they would have gone off their collective rockers in the first three weeks.

One by one, Roysland Dwyn called in the best analysts from every sector of the human-controlled galaxy. And slowly the information began to build up.

The first firing test of the enemy weapon was conducted on Syndor, the outermost and smaller of the two satellites of Kandoris VI. Roysland had the thing taken to the subnucleonics lab there because he felt that there was no need to subject the population of Kandoris to any danger from the backwash—if any. And only God knew how much territory the effective field might cover.

The Special Weapons group had dismantled one of the projectors from the ship and loaded it carefully on the X-69, along with the Enlissa-built generator that powered it.

On Syndor, Roysland watched the unloading. He stood on the broad, airless stretch of the landing field and watched the grapples lower the big, tubular weapon to the deck of the field. The blue-white glare of the distant sun splashed off the metallic sides of the ship, forcing Roysland to narrow his eyes, in spite of the heavy polarized filter in the helmet of his spacesuit.

The thing floated down under the control of the grapple beams until it was only a few feet from the surface.

Roysland heard the voice of the crew leader bellow in his earphones. "O.K., watch it! Get the truck underneath that thing before you drop it any more!"

A sturdy six-wheeled truck was moved in under the projector. The grapple operator turned a rheostat, and the projector sank another six inches, to rest on the truck.

"O.K.!" yelled the crew leader. "Haul her away!"

The truck trundled off in the direction of the Llordis Mountains.

Kiffer's voice came through Roysland's phones. "Let's go, Roysland; I'm right behind you."

Roysland turned around. Kiffer Samm was sitting in the driver's seat of a small jeep.

As he climbed in, Roysland said, "I felt the vibration as you pulled up, but I didn't pay any attention to it. Coming up behind a guy like that is real sneaky."

Kiffer's chuckle coincided with the slight vibration of the jeep as it started moving after the six-wheeler.

The testing area was some miles from the permanent labs. Roysland wanted to test the weapon by firing at Kandoris herself. The huge blue-white sun could certainly take anything directed at her.

It took the better part of three days to set up the site for the test, and during most of that time, Roysland Dwyn was in a spacesuit. The construction engineers had rigged up a plastic shell for dormitories and other inside necessities, but the work had to be done in the vacuum of space. By the time the set-up had been completed, Roysland felt exhausted in every muscle of his huge body. On the "afternoon" of the third day, he peeled off his oversize spacesuit and lay back on his cot. It was much too short for him, and his feet stuck out over the edge; but he was too tired to worry about that.

Kiffer was sitting on his own bunk, massaging his neck to get the kinks out. "The thing that bothers me," he said, "is the eternal sunlight. That blasted star won't go down for another seventy days."

Roysland nodded, but it was obvious that his mind was elsewhere.

"Suppose there is a backwash from this thing," Roysland said at last. "That would account for a lot of things. We've been wondering why the Enlissa ships didn't loot our own vessels after they used the mindjammer."

"Certainly," Kiffer said. "It's obvious. Their own weapon backfired on them, and left the Enlissa ship incapable of doing any looting. I figured that out a long time ago."

"Oh, did you?" asked Roysland smoothly. "Then did you figure out why the Enlissa didn't test the thing before they used it?"

Kiffer shrugged. "Who knows? What do I know about alien psychology?"

"You don't have to know anything about psychology of any kind; all you have to know is a common, ordinary law of species survival. Any race that takes a weapon into battle without testing it thoroughly, doesn't survive very long."

Kiffer ran the tips of his fingers across his lower lips. "True; but maybe they were suicide squads—or maybe they have a hospital ship following them to pick them up and cure them. After all, Bilford has this cure of his working pretty well now; if the Enlissa invented this thing, they probably know how to counter its effects.

"Besides, you didn't think we'd tested the *aJ* guns thoroughly. And we're still surviving."

Roysland turned to look at Kiffer, and his face was definitely sneering. "Kiffer, there are times when your thinking has all the clarity and lucidness of a hunk of obsidian.

"There's a difference between the lack of testing of the *aJ* gun and the Enlissa's not thoroughly testing the mindjammer. There's a difference between looking for something you could logically expect and not finding something that you don't even suspect the existence of."

Kiffer nodded. "Sure; I see what you mean. But that simply means that they don't have any way of shielding the effect—so they have a hospital ship trailing them."

Roysland lay back again and closed his eyes. "Obsidian," he said. Then, after a moment, "One: Why do they sacrifice a crew—even if it's only for a short time? Two: Why don't they use such an efficient weapon against ships that blast them out of the sky? Three: Why do they come in at a ship without firing anything at all?

"Until your hypothesis answers all of those questions—and a lot more besides—it isn't worth a damn."

Kiffer chewed at his upper lip and then looked at his wrist watch. "If you're going to test that thing in an hour, you'd better call Eckisster now."

Roysland sighed deeply. "O.K.; I'll call Old Nasty. Give me a minute to brace myself."

He didn't take the minute; he didn't really need it. He walked over to the solidiphone and punched in the code numbers. Three seconds later, General Director Eckisster was sitting in the middle of the room.

"You're ready, eh? All right; go ahead," he said. "Find out what you can—if anything. I have no further instructions—just don't get yourself killed while you're working."

The heavy space boot that came from Roysland's hand sailed through the image just as it was dissolving. Eckisster had cut off without waiting for Roysland's answer.

"One of these days," Kiffer said, "you're going to be in his office, and you'll forget it isn't a solidograph image and let go with a boot, or something, and knock the boss' teeth in."

Roysland shook his head emphatically as he walked over to pick up the boot. "Nope. If he's actually there in person, I'm going to have a poisoned needle to jab into him. I'll show him how to needle people!"

The Enlissa weapon was fired at Kandoris at 30:00 hours. Spaceships posted along the long line of fire between the satellite of Kandoris VI and the sun itself had sent out instrument-filled drones in the path of the beam to check the beam frequency. The time required for the subetheric wave to travel the eight hundred million miles from the planetary orbit to primary was too short to be measured. As far as the recording instruments were concerned, the beam was instantaneous.

The projector itself was fired by remote control; there were no personnel within three miles of the Enlissa projector when it went off.

The resultant recordings were run through the differential analyzers, and the final graphs were delivered to Kiffer Samm.

After four hours of working with the data, Kiffer made his report to Roysland.

"It's an odd wave length," he said. "Actually, it's a harmonic of three different basic frequencies. Look here: the thing is definitely frequency modulated, but it's a comparatively simple thing." He ran his finger along the primary recordings. "The thing wouldn't really have to be run through the differentials; it could be figured out with a slipdisk.

"The thing that makes it different is the extremely short wave length. The longest of the three has a wave length of eighty thousand kilometers, and the shortest is forty-two thousand kilometers. In a subetheric beam, that's the equivalent of hard X-rays—damned high frequency."

Roysland looked at the recordings carefully. "Is there any reason why this particular wave length should have any effect on the human brain?"

Kiffer looked at the graphs for a long time. When he finally looked up, he said: "I don't know for sure; mind if I call Bilford?"

Roysland shook his great head. "Go ahead; I don't mind."

When Bilford's image flickered into existence, Roysland kept his mouth shut while Kiffer showed the psychometrist the recordings of the energy from the Enlissa projector.

Bilford listened and looked and frowned. "The recordings actually don't make sense to me," he admitted. "I'm a psychometrist, not a subelectronocist.

"If you could translate those recordings from subetheric to their electromagnetic equivalents, I might be able to make something out of it."

The conversion didn't take long; all Kiffer had to do was run the stuff through the analyzer and punch in a correction factor.

Bilford stared at the corrected graphs and compared them with tracings of his own.

"I don't see any correlation," he said at last. "This may take a bit of work. There may be multiple harmonics of the basic stuff involved, of course; but frankly I can't see that the subetherics have anything in common with the electromagnetics as far as this area is concerned."

For the first time, Roysland spoke. "Try a combination-permutation synthesis. See what you get—O.K.?"

Bilford nodded in agreement. "I'll try it—all the different wave lengths involved, plus the subetheric velocity factor. If I come up with anything, I'll let you know."

"Good enough," said Roysland.

The solidiphone image of General Director Eckisster stood in the center of the room. He looked around and then focused his gaze on Roysland Dwyn. "Listen here, Roysland," he said belligerently, "why haven't you done anything? What's the situation now?"

Roysland looked at the general director and put on his nastiest grin. "You've got the report; we haven't done anything. We've fired the Enlissa projector six times. There is only a residual backwash that is harmless. You could fire

the thing in your living room if you wanted to. Meanwhile, we want to know what the effect of the beam is."

"And why, may I ask," said Eckisster, "can't you determine so simple a thing as that? This request is utter and absolute nonsense!" He slapped at the papers he held in his hand.

"I knew you'd like that," Roysland said. "I thought maybe you could suggest something else. I can't."

"As I understand it," Eckisster said testily, "you want a human volunteer to test the Enlissa mindjammer on."

"That's right," Roysland said. "So far, all we've proven is that the backwash from the projector has no effect on humans or animals; but we don't know what happens to a man who's hit by the beam itself."

"Oh? We don't? I rather assumed that the fleet hospital's psychiatric wards were full of men who have been hit by the beam."

"An unjustified assumption," Roysland snapped. "At least, so far, it's unprovable. The point is: Do I or don't I have your permission to ask for a volunteer?"

"Why can't you use test animals?" Eckisster asked.

"If you'd bother to read the reports I send you, you'd know. We *have* used 'em. The beam didn't touch 'em. We sprayed one group for half an hour; and as far as anyone can tell, we might just as well have been shining a flashlight on them."

"Of course," Eckisster said. "The mindjammer causes a feedback loop in the prefrontal lobes. What do you expect it to do to animals with no prefrontal lobe?"

"My point exactly," Roysland agreed. He knew perfectly well that Eckisster had read the report completely and thoroughly. His pretended ignorance and snide remarks were just a mechanism he used for purposes of his own.

"The question is," Roysland repeated, "do I have your permission to ask for a volunteer?"

"I checked with Bilford," the general director said. "He's getting the microwave technique worked out fairly well now; he says he can bring a

man around in twenty-five to thirty days." He stopped and looked at Roysland closely. "Go ahead and ask for volunteers."

"Thanks," said Roysland.

Eckisster nodded as he dissolved.

Roysland reached over and punched a button. "Where's Kiffer?" he asked.

"Eating at the mess hall, right now," said a voice.

"That's what I thought. Will you have him come here, to my place, as soon as he gets through? Say, in half an hour?"

"I'll tell him."

"Fine." Roysland lifted his finger and turned to the typer on his desk. He wasn't used to the makeshift office, and he found himself wishing he was back on Kandoris VI, in his own office.

He shrugged and began running his fingers over the typer. It took him only a few minutes to put down what he wanted to say. When he finished, he pulled the sheet from the printer tank and put it on his desk, in plain sight. At the top, he scrawled: "To Kiffer Samm." His own signature went at the bottom.

Then he put on his spacesuit and headed out, toward the outside air lock.

Half an hour later, Kiffer Samm was reading the note. He had stepped into Roysland's office and seen that it was empty. Assuming that his superior would be right back, he had sat down to wait. Then he'd seen the note.

He was halfway through it before it became perfectly clear what Roysland was doing.

"... So you may have to take over for the next twenty-five to thirty days. Naturally, I couldn't ask anyone else to take the risk.

"I think it may be a good idea if Bilford starts experimenting with subetherics in an effort to snap the rest of the boys out of this feedback loop thing. Maybe he can do it in less time.

"By the time you read this and get in a spacesuit and get out to the firing area, I will have finished the test; don't let me die of starvation, chum."

Kiffer punched at the communicator button, yelled orders into it, and grabbed a spacesuit out of the locker. By the time he reached the outer air lock, a jeep was waiting for him.

When the second jeep pulled up, Kiffer said: "You men stop at the gun emplacement and take a look at the weapon. We'll go on to the target tower and pick up Roysland."

The men nodded their agreement, and the two vehicles started rolling.

Theoretically, it was "evening," but the great, blue-white blaze of Kandoris still hung in the eternally black sky. The jeep went by the gun emplacement where the Enlissa weapon had been set up for testing. Kiffer noticed that the snout of the ugly-looking tube was aimed at the squat steel tower where the animal subjects had been exposed to its radiations.

"There he is!" said the jeep's driver, pointing.

Kiffer could see a spacesuited figure on the target tower. He twisted the dial on his chest and said to the men in the second jeep: "Check that projector! Make sure it isn't in operation!"

"It's not," said one of the men. "He had a timer connected to the firing mechanism. He got a ten-second burst from it, according to the timer reading."

"Thanks. We'll pick him up, then."

The jeep swerved toward the tower and pulled up underneath it in a swirl of dust that settled slowly and evenly in the low gravity of the airless satellite. Kiffer jumped out of the jeep, grabbed the rungs of the ladder, and lifted himself to the platform at the top of the twenty-foot tower.

He stuck his head up over the edge and saw Roysland. The man was sitting on a small chair with his back to the ladder. Surrounding him were the various recording instruments that had been rigged up on the platform for testing the animals and the effects of the beam on them.



Kiffer climbed on up and twisted his helmet phone control to Roysland's frequency. As he put his hand on Roysland's shoulder, he said: "Stand up, Roysland."

Roysland jerked around. "What? Oh. Hi, Kiffer; I saw you coming in the jeep." He paused then, and though Kiffer couldn't see very well through the heavy darkness of the helmet's glare-filtering polarization, he could have sworn that Roysland was grinning. He would have been right.

"Oh, I get it," Roysland said. "You were expecting to find me sitting up here with a feedback lobotomy. Frankly, so was I, a half hour or so ago, but I'd almost forgotten it."

Kiffer took a deep breath, let it out, and said a few choice, pungent words. "... Who would scare a guy like that," he ended.

"Sorry," Roysland said, still grinning. "But take a look at these readings. I think you'll—"

"Wait a minute!" Kiffer interrupted. "I'm not interested in meter readings right now! What happened or didn't happen to you?"

"Is he all right?"

"What's going on up there?"

"Need any help?"

The voices came almost simultaneously to Kiffer's phones. He could see the second jeep tearing up dust between the gun emplacement and the target tower.

"He's O.K.," Kiffer snapped. "Big false alarm! I think we ought to have an explanation."

The answering burst of catcalling and jeers made Roysland wince. "O.K., fellers! O.K.! Please accept my abject and snivelling apologies."

"Explain yourself," Kiffer said in a monarchical tone. "You were supposed to be out here testing this thing on yourself; you wrote a very heart-rending note to that effect. I don't blame you for getting cold feet, but you could at least have notified us."

"I didn't get cold feet," Roysland said. "Look at the cerebrograph reading and compare it with the firing record."

Kiffer looked and then said: "Then you *did* take it! But according to this, all it did was cause a very faint *petit mal* convulsion. You probably didn't even notice it."

"I didn't," Roysland said. "I don't know what that projector is supposed to do, but it sure isn't a mindjammer!"

Kiffer looked again at the records. "Maybe you weren't far enough away from the projector," he said doubtfully. "Maybe the distance—"

"Impossible," said Roysland. "The beam doesn't disperse appreciably over a distance of half a light-year; you know that. And the wave form is exactly the same."

"No, I'm afraid we've just run up against another blind alley."

Kiffer shook his head slowly. "I don't believe it," he said. "The Enlissa didn't have their ship armed with this thing for nothing. We must have connected it up wrong, somehow."

"Maybe," Roysland said. "But it doesn't work as is. Let's get these records into the jeep; I want to see what we're getting here, anyway."

They took the recordings out of the instruments and dropped them to the three men who were waiting by the jeeps parked underneath the tower.

A few minutes later, they were heading back toward the dome.

Four days later, Roysland was back on Kandoris VI, ensconced firmly in his office. Kiffer Samm stayed on Syndor, still working on the Enlissa projector.

The first thing Roysland did was to call another staff meeting. He also included Bilford and Commander Allergyce.

He outlined briefly the data they had so far on the Enlissa mindjammer, then asked for comments.

Bilford grabbed the floor first. "I did the correlation you wanted, and I came up with some answers, but they're not the right ones as far as I can tell.

"As far as the backwash on the *aJ* gun is concerned, I think you can rule that out. After converting to electromagnetic equivalents, I find that the frequency of the backwash is much too low to have any effect on the brain. That is, assuming that subetherics have any effect on the mind at all—and, of course, assuming that there is any analogy at all between the function of subetheric vibrations and electromagnetic vibrations. After all, analogue reasoning has its limitations, too, just as logical reasoning does.

"The captured Enlissa projector is another problem. Unlike the *aJ*'s backwash, it isn't a noise; it's a definite, although complex, tone. I say complex because—and again my reasoning is analogical—because the frequency is not a pure sine wave, but a combination. It's analogous to the difference between the vibration of a tuning fork sounding middle C and, say, a violin sounding the same note.

"Even so, I think we can say that the captured projector is *not* the mindjammer; the frequency is much too high. It's on the order of hard X-rays. If the analogy holds, the subetheric beam should be capable of disrupting certain molecules, but it most certainly couldn't have the mindjamming effect on the human brain."

He sat down and rubbed his hands together nervously.

Commander Allerdyce stood up. Normally, the fleet commander did not kowtow to anyone, but his automatic respect for the big man in the chair at the head of the table came to the fore. As a matter of fact, the commander didn't think of it as kowtowing; he merely acknowledged the superior abilities of the man he was facing.

"All I've got is statistics, Roysland. I wouldn't have noticed it without your hint, but we've worked out a new strategy that has reduced casualties by better than sixty per cent." He reached down and picked up a pile of report sheets.

"It stacks up this way: About thirty per cent of the Enlissa ships that attack have the habit of coming in without firing anything. What the reason is, I don't know, but they do it. Therefore, we have a good chance of getting the enemy with torpedoes alone if he doesn't fire first.

"A ship equipped with *aJ* projectors has about a seventy per cent chance of winning. The other thirty-odd per cent of the time, they're mindjammed.

"The chances of a conventionally armed ship coming through is better than sixty-two per cent.

"But here's the gimmick: In taking the action of the Enlissa fleet into account, we can reduce the casualties tremendously. About thirty-two per cent of them come in without firing. By taking that into account, we can increase our own chances of survival tremendously."

Roysland nodded. "Good; I'd like to see the statistics on that. Would you mind sending over the full report?"

"Not at all," said Commander Allerdyce. He sat down.

Taddibol stood. "I think I can speak for Vanisson, Mardis, and myself. According to the evidence we have, the Enlissa are capable of picking out a ship with *aJ* guns *before* they fire. We think that there must be some residual

emanation from the *aJ* that is detectable by the enemy. No other hypothesis fits the facts."

Vanisson was standing before Taddibol had finished. "I'd like to make it clear that, although I agree with Taddibol Vlys, the evidence is still a necessary part of the hypothesis. We've—"

The emergency buzzer sounded, and everyone at the table turned to look at Roysland as he swore roundly and jabbed the stud. General Director Eckisster had barely begun to solidify before Roysland said: "Can't I have any peace? Must you continually and forever be looking over my shoulder?"

"No," said Eckisster calmly. "Yes. If that answers your questions, may I say something? I'm sorry I had to interrupt a staff meeting, but I felt that this would be the perfect time to inject this bit of data.

"As I see it, you weren't satisfied with human volunteers for the Enlissa weapon; you asked that two of the aliens also be subjected to the beam from their own gun."

"That's right," Roysland said. "According to Bilford, two of them have been rendered sane by the treatment of the microwave frequencies. I didn't think you'd reject using the Enlissa captives on humanitarian grounds."

"I didn't," Eckisster said. "Your man, Kiffer, claimed that further information could be gained by subjecting the alien brains of the enemy to the radiation from their own projector. Since the psychological department has now discovered a method of bringing back the functional ability of the brain after exposure to the mindjammer effect, I didn't think it would be harmful to allow two of the aliens to be subjected to it again. Unfortunately, they died."

"They *what*?" Bilford shouted the question.

"Died, Bilford, *died*," Eckisster said. "They are both as dead as the surface of Syndor."

"Good God!" Bilford said. "Perhaps a second exposure—" Suddenly he jammed a finger down on his cutoff, and his image vanished from the conference room.

"What was the reason for that?" Eckisster wanted to know.

"He's just released the first batch of men from the hospital for active duty," said Fleet Commander Allerdyce. "If that thing *is* the mindjammer, and those

men are exposed again—Excuse me." His own finger touched the cutoff, and his image flickered out.

Eckisster looked at Roysland. "Well, sir?"

Roysland shook his head. "I didn't expect that," he said. "I honestly didn't expect that."

"I know you didn't," Eckisster said softly. "I know you didn't. But look at it this way: It's data. And we need data."

"I know," Roysland said. "It's not that. Excuse me; I've got to think." He slammed his hand down, and the whole group collapsed into nothingness.

"What?" asked Commander Allergyce.

"I said," Roysland repeated, "that I think I have the answer to something that was brought up in the meeting last night. And I want you to give me permission to take the X-69 into enemy territory."

"I will," Allergyce said, "if you'll give me a good reason for going."

"All I want is a sample of alien animal life. I think I know what's going on, but I'm not sure."

Allergyce shook his head. "We can't do it. We don't know where the enemy bases are, any more than the Enlissa know where our own planets are. We keep our subetheric devices shielded, and so do they. If we didn't, this would have ceased to be a spatial war long ago—you know that."

"I know," Roysland admitted; "but we have prisoners; members of the enemy's armed forces. We can get our information from them."

Allergyce was still shaking his head. "How? They've been treated mentally against probing. They won't tell us where their home planets are, any more than our own men would—or *could*—tell them."

Roysland, in turn, shook his head. "That's not what I'm looking for. I'm not a military man; I'm a scientist—at least I think I am. I'm not looking for military bases; I'm looking for a planet where the Enlissa have planted their flora and fauna. That's what we do with a planet, isn't it? Seed it long before

we colonize. If they've done as much colonization as we have—and their war potential shows that they must have—then they'll have a lot of planets that aren't inhabited by the Enlissa themselves, but will have been seeded by Enlissa-type life.

"At least one of the crewmen from that ship will know where such a planet is located. And I'm willing to bet that he won't be conditioned against telling us."

"Why not?" Allerdyce asked.

"For the same reason you haven't thought of it," Roysland said, grinning. "The Colonization Service and the Fleet Command are two different branches. Unless the aliens think differently than we do, their organization is about the same. And every bit of evidence shows that their reasoning is similar.

"There's no reason to protect an unpopulated planet, is there? Besides, the military don't inspect colonization records. Why should they? And what would it matter if the enemy took over an unpopulated planet? After all, we have as much chance of taking over one of theirs."

Allerdyce thought it over before answering. Finally, he said: "I'll check with Bilford. If he thinks we can get that much information out of an alien, I'll O.K. the trip. I'll have to insist, of course, that the X-69 be fully armed and subject to military orders."

"Naturally," Roysland agreed. "Just let me make the trip; that's all."

"I'll see what I can do," said Allerdyce. "Meanwhile, I'm going to call Colonization Service."

Roysland smiled to himself as he cut the connection.

Three days after that, the X-69 lifted again for space. On board her, locked securely in the brig, was the first officer of the captured Enlissa vessel.

No one had yet determined the nature of the Enlissa language, but Bilford had worked out a method of getting yes-no answers out of him, and had, by the process of elimination, arrived at a star system that contained a planet

which had been seeded by the aliens. And all Roysland wanted was a sample of the Enlissa animals.

There's an old saying which goes: "Some people have all the luck." It has echoed down the corridors of human history and human thought for a thousand centuries, in one form or another. It is usually assumed to be the complaint of the unsuccessful against those whose success is greater—but it is to be noted that it is not specified whether the luck is good or bad.

With the same reservations, one might assume that Roysland Dwyn was lucky. On the fourth day out, the alarm buzzers sang their warning through the corridors of the X-69.

As the crew scrambled for battle stations, Roysland headed up the stairway toward the control bridge. Captain Dobrin and the fire control officer were huddled over the spotterscope, conversing in low tones. Roysland walked over behind them, but he kept his mouth shut. In a situation like this, he was only a civilian; it wasn't his business to say anything now. He studied the instruments, instead.

Somewhere out near the limits of the detector's range had come the faint trace of a moving ship. And the identity comparator showed it to be an Enlissa vessel.

"She must have picked us up, too," said the captain. "We'll know in a few minutes."

They watched quietly, tensely, waiting for the Enlissa ship to change course. If it didn't, a battleship would normally change the geodesic of its own flight and follow to engage the Enlissa ship. But not the X-69; she was looking for planets, not ships.

They didn't have to wait long. A few minutes later, a trace appeared in the same octant of the scope where the earlier trace had vanished.

"Same ship, all right," said the FCO. "It would take them that long to turn around. They're going to try to come in for a kill."

"Signal Final Alert," said the captain.

As the buzzer sizzled out its message, Roysland flexed his muscles in a subconscious desire for action.

Captain Dobrin seemed to realize for the first time that Roysland was in the control room. His face was hard and tightly drawn, and only very slightly showed the strain that was beneath.

"We're going to operate according to the new tactics," he said. "We'll use the torpedoes first and the *aJ* guns last. We'll use screenbusters and files."

Roysland nodded. "You're in command here, captain. I know nothing of spatial strategy."

The prime officer turned back to the FCO. "Check maximum volume and englobe. It'll be expensive, but we can't afford to take chances now."

"Yes, sir," said the fire control officer.

Roysland watched the instruments closely as the FCO gave his orders. The first job was to feed into the calculators the exact course and velocity of the enemy ship. Then they waited until the calculators gave the most probable volume of space that the ship would occupy after the screenbuster torpedoes were sent.

Take a solution to the Brownian Movement problem; add everything that is known about spatial strategy; stir well with the enemy's probable interpretation of the signals from the torpedoes—and hope like hell.

The first ingredient is relatively easy to determine, the second very much less so, and the third is almost pure intuition.

Figures began to pop up on the screen. The FCO watched them, unmoving, his face a rigid mask. Then suddenly, he began to punch data into the torpedo-firing robots.

Roysland narrowed his eyes as he watched. The *aJ* projectors didn't require that much computation. If the *aJ*'s were fired now, the Enlissa ship wouldn't have a chance to fire. And yet, statistics showed that—

Why?

The FCO's masklike face began to acquire a sheen of perspiration in the glowing lights of the control room as he watched the screen and punched methodically at the fire control board. It was work that no robot could do; it required the shrewdness, intuition, and foresightedness that is a peculiar quality of the human mind.

Without warning, the FCO jabbed violently at the white stud that stood at the edge of his panel. He jerked his finger off, and his hand seemed to freeze for a second. He had done the irrevocable; he had fired every torpedo in the ship.

The X-69 now possessed no armament except the *aJ* guns.

The first volley of screenbusters left the ship and slammed suddenly into the ultravelocity that only an unmanned torpedo is capable of. Even an antiacceleration field isn't one hundred per cent perfect. In no-space drive, a ship can accelerate at the spatial equivalent of better than a hundred thousand gravities without hurting the crew. But the tremendous acceleration of a war torpedo would crush any human body to a monomolecular film.

The torpedoes had to be small; only a very small no-space generator could achieve such velocities in so short a time. But their small capacity was capable of carrying enough subnuclear explosive to smash through the energy screens of the enemy ship.

They could not, however, breach the vodium hull itself or kill personnel within. That was where the "flies" came in. Their job was to smash through the breach in the energy screen, open the hull, and destroy life within.

The only trouble was that the enemy could detect the torpedoes. If the Enlissa could act fast enough, they might be able to avoid them. The hope of the human ship was that the englobement would be too much for the robot computers of the Enlissa.

The first wave of torpedoes left the X-69, spearing in the general direction of the Enlissa vessel. For a fraction of a second, they maintained their original course. Then they became erratic—purposely so. They flashed on and off in the detector screens as their no-space generators cut in and out, and they switched courses with dizzying rapidity. They had been on their way only for hundredths of a second when the second volley let go. Then the third blasted out. The whole thing was over before an eyelid could flicker.

Roysland glanced at the chronometer; the whole operation had taken slightly over ninety seconds.

The silence lasted only for a moment. One of the observers called out: "Torpedo at twelve thirty-seven!"

The data had already been picked up by the robot pilot, and the X-69 shifted course. Roysland could feel the slightly sickish feeling in his stomach under the heavy acceleration as the angular acceleration of the ship changed.

There was nothing to do now but wait. It was up to the robot defenses and the screens to make sure that no enemy torpedo hit the X-69.

The ship lurched again.

Because of their tremendous acceleration, a war torpedo couldn't possibly be a homing type weapon; it moved too fast. Before even a subelectronic relay could operate, the target would be well out of range. The X-69 was in the position of a man ducking thrown stones; the only fatal move would be an inaccurate judgment.

Again the floor jerked beneath them as another enemy torpedo sizzled through the place where the ship might have been.

"Explosion at fifteen-sixty!" shouted two observers at once.

The FCO's face suddenly broke into a grin. "We did it," he said softly.

Then the intercom flickered on. An excited Space Marine said: "Captain Dobrin! There's something funny going on down here; that Enlissa officer we've got in the brig just dropped dead!"

It was at that instant that Roysland Dwyn found his answer. The pieces of the whole jigsaw puzzle fell into place and made a beautiful picture. And he realized that the Enlissa, too, had changed their battle tactics.

And that was when the explosion hit.

Four torpedoes had come in on the X-69 at once, and the robot had been a fraction of a second too late in computing the trajectories all at once and figuring a safe path.

The screenbuster's detonation jarred the whole ship violently. Then there were two thumps as a pair of flies came into the hole through the screen and

blasted the interior of the cruiser.

Roysland wasn't sure what had happened; the whole control room had suddenly seemed to turn upside down. When he picked himself up from one wall—which had now become "down"—his nose was bleeding, and his right arm was dead to the shoulder. Broken clavicle.

He shook his head groggily and looked around. Captain Dobrin was slumped against a corner of the wall. The FCO was sprawled across the side of his control board. The various observers were tumbled around the room like so many rag dolls shaken up in a shoe box.

Gradually, the gravity righted itself, and Roysland rolled to the floor. He pulled himself up by one arm and ran toward the control panel. He had barely time to act.

Fortunately, most of the observers were reasonably aware of their surroundings. Those who could move were back at their control boards by the time Roysland reached the fire control board.

A second blast hit the ship, but Roysland was prepared for it this time; his fingers gripped the hand-holds and strained as the gravity shifted beneath his feet.



The X-69 couldn't stand another one like that. The Enlissa ship had computed better than they had thought.

"*aJ* projectors!" Roysland shouted. "Prepare to track and fire!"

The only way to save the ship now was to shoot down every torpedo before it hit.

"All guns tracking, sir," said one of the observers.

"Set and ready!" Roysland said. "Fire automatically!" He punched a button.

The *aJ* projectors moved in their mounts, each one seeking out a different missile. They would go on seeking until the—

Then the first one fired, and Roysland's mind went blank, as did everyone else's aboard the X-69.

For a long time, Roysland Dwyn watched a play. He was a disinterested spectator, who had not one iota of interest in what was going on. He was much, much, *much* too busy with his own thoughts to be interested with such trivia as his bodily reactions and his exterior environment.

In the first place, he had solved the problem. And such a fascinating problem! The broad ramifications of the whole concept were appalling in their immensity and scope!

Some people came into the control room after a long while and asked him some questions. He answered them politely, but without paying any attention whatsoever to what they were saying.

After all, what could possibly be so utterly absorbing as my own problems? Who could be more important than I?

The people asked him to walk to somewhere, and he did; but he didn't have the slightest notion where he was going, nor why, nor how. And he really didn't care. They put him in a bed and fed him soup and stuck needles in his arm and several other utterly meaningless things, but it made no difference.

Introspection. Know thyself. And then get going around and around and around on the ever deepening spiral-helix that goes lower and lower as it closes in on itself. Self-analysis. What are my motivations? Why do I want to know what my motivations are? Why am I analyzing myself? Why do I want to know why I am analyzing myself?

What do I know about the motivations for desiring to know about the reasons for analyzing myself? Why do I feel that the motivations—

After a long period of being left alone, he was in a place that was different from where he had been before, but it wasn't any different than the place where—

A sudden blazing shock crossed Roysland's mind. With the awful brilliant clarity of a man seeing suddenly into a darkened room when the lights have been lit unexpectedly, Roysland snapped agonizingly back to awareness.

Only for a fraction of a second did he realize what had happened. Then his mind blacked out under the shock.

When he came out of it again, a nurse was standing by his bedside. She smiled at him when he opened his eyes, and said: "How do you feel, sir?"

He thought for a moment, taking inventory of exactly how he did feel. Then he smiled. "I feel fine. What happened?"

The girl touched a relay plate. "The psychometrist will be in right away, sir. He'll explain things to you." She gave him another flash of white teeth and stepped out of the room.

Less than a minute later, the door opened, and the psychometrist came in. It was Bilford.

"Well, well," Roysland said. "I get special treatment; the chief cheese is in to see me."

Bilford grinned, ran a hand through his hair and nodded. His thin face seemed to almost sparkle from within. "Yup. You're important. I knew you'd want to see someone as soon as you came to."

Roysland propped himself up in bed. "How right you are. The boys have solved the Secret of the Mysterious Weapon, I see. Have they actually made a usable weapon out of it?"

Bilford lifted his eyebrows. "What makes you think they've figured it out?"

Roysland's massive face broke into a grin. "Simple. I'm back among the living again. If I'm right—and I think I am—you undid this feedback in the prefrontal lobes with an effect similar to the one that caused it. Q.E.D.: You know what caused it."

Bilford nodded. "Good reasoning. And accurate. I guess your brain isn't as burned out as it might be. I guess you can see visitors now."

"Who?" Roysland asked.

Bilford stood up and headed for the door. "Four Special Weapons staff members and a Fleet Commander. They've been waiting to see you for three days, and I told them you'd be out from under this morning." Then he stopped at the door and looked bland. "Of course, if you don't *want* to see them—"

"Get them in here!" bellowed Roysland.

All Bilford had to do was open the door. Five men came into the room as though the hall were full of poison gas. After a minute or so of inquiring after Roysland's health and expressing their sympathy for his plight, they settled down to business.

"I figured there was something screwy in that story you gave me," Allerdyce said. "Going to hunt for animals, indeed!"

Bilford grinned. "I didn't think he was, either. It was brilliant to have those recorders in the Enlissa officer's cell. And the other stuff came through perfectly."

Roysland shook his head. "You misunderstand me. I most certainly did intend to get animal specimens. I figured the answer was involved with the aliens themselves, but I didn't know what the gimmick was.

"Now I know that it was the interaction of the *aJ*'s backwash and the enemy's beam that caused the mindjammer effect. The enemy's weapon was intended as a death ray, but for some reason, it doesn't work on humans."

"That's right," said Taddibol. "The enemy projector was designed to disintegrate the molecule of a particular enzyme that is necessary to Enlissa life. It does the job beautifully, too. When the beam hits an Enlissa, the enzyme disintegrates, oxidation can no longer take place in the tissues, and presto! the Enlissa dies. But our own system is so different that the beam doesn't even effect us."

"The answer's been right in front of our eyes for a long time," Kiffer said. "The backwash from the *aJ*'s has too long a wave length to be effective, and the Enlissa's death ray is too short. But the complex harmonic of the two is just right. It creates a momentary field that causes the loop-feedback to start in the prefrontal lobes. From what we can gather, the effect is one of intense, overpowering curiosity—inwardly directed."

"Statistically," Allerdyce cut in, "it accounts for the peculiar behavior of the enemy ships, too. If we assume that a little over twenty-five per cent of their ships are equipped with what they think is a death ray, you'll get the right figures. About the same number of our ships are equipped with *aJ* projectors.

"When a death-ray ship comes in on an *aJ* ship, the *aJ* guns cut it down and the crew is mindjammed. But if a death-ray ship comes in on one of our conventionally armed ships, they're blasted out of the sky because they

figure that everyone aboard the ship is dead and they don't bother to fire any torpedoes. Our own torpedoes come as a pretty rude surprise. So the enemy has lost one hundred per cent of their death-ray equipped vessels in every engagement!"

Roysland nodded. "We couldn't see it because we weren't looking for it. I suspected at first that it had something to do with the *aJ*'s; the statistics suggested that. But when every test showed that it couldn't possibly be our own projectors, and when this Enlissa projector came along, I made the mistake of dropping the previous line of approach. Keep that in mind, boys; you can forget old *theories*, but you can't forget old *data*.

"By the way, commander, did you figure out how we happened to get the Enlissa ship?"

"Sure," said Allerdyce. "When they came in so close, they were caught by the field that was generated. The thing has an effective englobement volume with a radius of about six hundred miles. We don't know what the effect is near the outside, of course, but we're working on it."

"You know," Roysland said, "mankind has known for centuries the old dictum that 'the whole is greater than the sum of its parts,' but we sometimes forget how it works in practice. We still tend to look from cause to effect and from effect to cause.

"But in this case, there were two 'causes' of the mindjammer field, and three 'effects' from the two 'causes.' And that's simplifying a great deal. We still haven't dug into what else we can get from subetheric harmonics phenomena."

Roysland looked at Bilford. "How did you do this quick-cure stunt?"

Bilford shrugged. "Simple. I fiddled around until I got a subetheric harmonic that corresponded to the frequencies of the microwaves I was using. Works fine."

Kiffer chimed in again with: "With the stuff we got from your instruments on the X-69 I think we can build the weapon we've been so afraid of."

"Won't the Enlissa be able to analyze it?" Bilford asked, interestedly. "After all, we figured it out."

"Not the same thing," said Kiffer. "They don't have *aJ* projectors yet. They can't accidentally generate the field."

"Besides," Commander Allerdyce said grimly, "we won't leave them any evidence. If the weapon works, we'll beam 'em down, board 'em, and end up with prisoners and a perfectly good ship. The Enlissa will never know what happened to them."

Roysland was about to say something when the door flew open and a heavy body propelled its way inside.

It was General Director Eckisster, and he was very obviously seething mad. He glanced around the room and his eyes lit on Bilford.

"May I ask, sir," he thundered, "why I have been kept from seeing Roysland Dwyn for two weeks? And why these men are allowed to see him now?" He didn't wait for an answer, but turned toward Roysland. "As for you, sir, I am filing a reprimand—officially. You had no business using the X-69 as military vessel during time of war without my permission. You might have been killed, and I need you!"

Roysland started to answer, but Commander Allerdyce was one jump ahead of him. He smiled serenely at Eckisster and said: "My dear director, don't you think such an action would be just a bit confusing? Captain Dobrin recommended that Roysland Dwyn be given the Golden Cluster for bravery in action above and beyond the call of duty. I added my recommendation and sent it on to the Regent's office. The Regent himself has given his approval. Surely, a reprimand now would be a bit unseemly."

Eckisster glowered. "My dear commander," he said, "it so happens that Roysland Dwyn is the mainstay of my directorate. It also happens to be a fact that I have a perfect right to threaten to do any damned thing I want to. It keeps him mad at me, so he works like a beaver to show me up. I threaten, cajole, intimidate, scream, and ask silly questions. It works. If you won't tell me how to run my directorate, I won't tell you how to run your spacefleet. At least not very often. Fair enough?"

Again, he did not pause for an answer, but looked back at Roysland. "And you, you get out of that bed as soon as this twitch doctor lets you. You have

a gun to build. A mindjammer. Get busy. I'll expect you in my office later. Good-by." He turned and stamped out.

Allerdyce stared at the closed door for a moment, then turned and grinned. "I guess I got told."

"You did," said Bilford, "and you're going to get told again. All of you. Clear out. The patient has had enough excitement for today. Scram."

It took the five men several more minutes to leave, but Bilford was finally alone with Roysland.

"Did you know that about Eckisster?" Bilford asked. "That he needles people with a purpose in mind?"

"Sure," said Roysland. "I've known it for years. I don't say that it works the way he thinks it does, but at least it keeps the job exciting. I think everybody needs a little needling now and then."

Bilford nodded. "I know you agree with him. You're a bigger needler than he is, any day."

"Me?" Roysland looked surprised.

"Yes, you. Eckisster's needling is effective in a limited way, but yours is not only effective, but efficient. You ask the kind of questions that make people think instead of the kind that make people mad. Where Eckisster jabs in all directions and people jump, you use your needle with the deftness and precision of a physician using a hypodermic. Eckisster doesn't know what he wants and he doesn't know how to get it. And he wants somebody else to do it for him, whatever it is. On the other hand, you know what you want and how to get it without making everybody hate you, and you'll do the job yourself, if necessary.

"You gave your staff men, Commander Allerdyce, even me, credit for finding out what the mindjammer effect was. But the credit belongs to you. If it weren't for your incessant needling, your ability to arouse interest in seemingly dull facts, your sometimes radical theories, and your propensity for asking searching questions, I doubt if we'd have our answer yet.

"The core of this problem wasn't just the fact that several phenomena combined to give the mindjammer; that was a purely physical effect. The big problem was to get human beings to take their individual fields of thought, work with them in relation to other fields of thought, and come up with useful information that could be fitted together to explain the whole.

"Eckisster's type of needling might make a man *work* harder, it might even make him *think* harder—but it won't make him think in a different way or look at data from a new angle. Even when your theories are wrong, you use them in such a way that they uncover the data which proves them wrong. And then you're perfectly willing to drop them and work out a new hypothesis and get people to try to destroy or confirm it." He stood up and smoothed a palm over his gray hair.

"And now, if you'll excuse me," he said, "I have some more things to work on. I have a hunch that these subelectronic polar harmonics can do a lot more to the human brain than just knock it silly. When you feel better, I'll tell you all about it." He turned and walked out the door.

Roysland lay back on his bed and looked at the ceiling. Me, a needler? he thought, ME?

THE END

*** END OF THE PROJECT GUTENBERG EBOOK NEEDLER ***

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